

August 26, 2026 03:36 AM GMT

Investor Presentation | Japan

Semiconductor Production Equipment:
Tech Monthly Aug 2026

MORGAN STANLEY MUFG SECURITIES CO., LTD. +

Suzune Tamura, CFA

Equity Analyst

Suzune.Tamura@morganstanleymufg.com

+81 3 6836-8891

Kazuo Yoshikawa, CFA

Equity Analyst

Kazuo.Yoshikawa@morganstanleymufg.com

+81 3 6836-8408

SEMICONDUCTOR PRODUCTION EQUIPMENT

Japan

Industry View

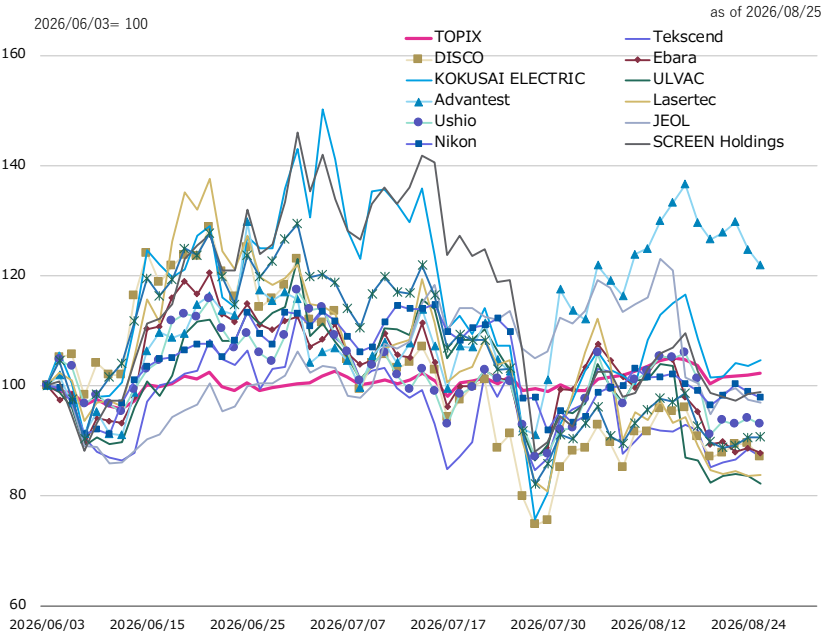
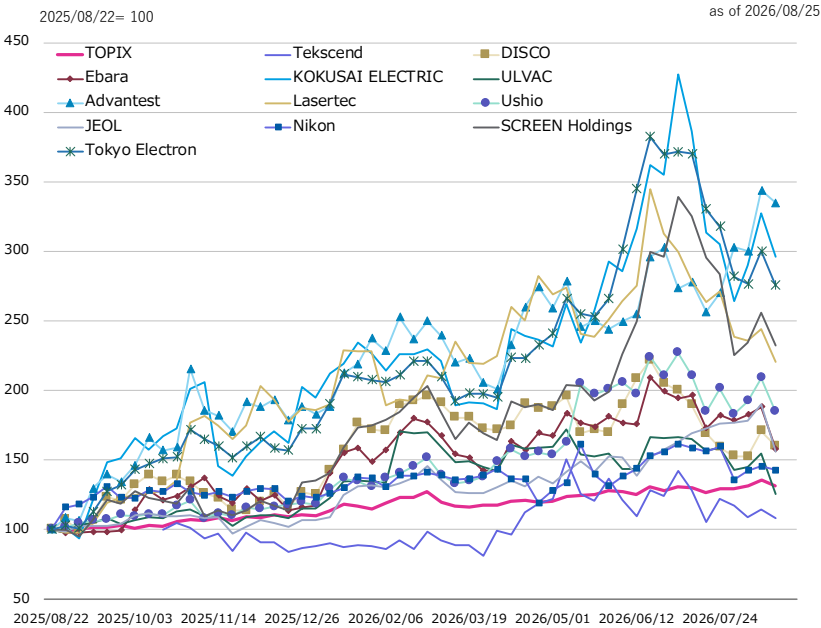
Attractive

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

+ = Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

Stock price performance



Source: FactSet data, Morgan Stanley Research

Stock performance and valuations

			Rating Share price							Valuation												
			(LC)		(LC) Performance			11/20 ~	Cons PE (x)			Cons EV/EBITDA (x)			Cons PBR (x)			Cons Div. yield				
			MS	Last close	MS PT	YTD	3m		1m	F1	F2	F3	F1	F2	F3	F1	F2	F3	F1	F2	F3	
Semiconductor Production Equipment (Japan)																						
8035 JP	Tokyo Electron	OW	55,410	65,000		61.5%	6.2%	-11.6%		70.5%	33.2	23.9	19.3	23.0	17.1	14.0	10.3	8.6	7.1	1.5%	2.0%	2.6%
7735 JP	SCREEN Holdings	OW	13,305	17,000		74.6%	20.6%	-16.1%		105.3%	21.5	16.6	14.0	13.1	10.5	9.1	4.5	3.8	3.2	1.4%	1.8%	2.2%
6525 JP	Kokusai Electric	OW	8,489	8,300		54.5%	11.0%	-0.9%		101.1%	36.0	25.3	19.4	21.1	15.7	12.5	7.8	6.4	5.3	0.7%	1.0%	1.3%
6146 JP	DISCO	OW	60,700	78,000		26.0%	-13.0%	-0.3%		28.4%	35.6	29.1	24.7	23.8	19.7	17.1	9.5	7.9	6.8	1.1%	1.4%	1.8%
6857 JP	Advantest	OW	34,450	36,000		75.5%	24.1%	19.0%		65.3%	38.1	29.8	24.8	28.5	22.6	19.1	20.5	13.8	9.9	0.3%	0.6%	0.8%
6361 JP	Ebara	OW	4,931	7,100		33.8%	-13.4%	-13.2%		27.7%	22.0	18.6	15.5	13.2	10.6	9.1	3.9	3.5	3.1	1.6%	1.9%	2.3%
6920 JP	Lasertec	UW	34,950	36,000		17.9%	-18.8%	-19.0%		25.0%	30.9	23.2	18.6	21.4	16.4	13.5	10.0	7.7	5.8	1.1%	1.4%	1.9%
6951 JP	JEOL	EW	7,140	7,100		41.9%	5.4%	-12.9%		53.8%	17.3	13.9	12.5	10.3	8.2	7.4	2.2	2.0	1.8	1.8%	2.3%	2.2%
6925 JP	Ushio	OW	3,900	4,900		55.7%	-7.5%	-6.3%		67.9%	29.7	19.9	16.5	12.6	9.9	8.2	1.5	1.5	1.4	1.8%	2.4%	3.1%
6728 JP	ULVAC	EW	7,597	8,100		7.3%	-21.7%	-21.0%		13.7%	17.5	14.9	13.7	7.6	6.6	6.2	1.5	1.4	1.3	2.1%	2.4%	2.6%
7731 JP	Nikon	UW	1,971	1,100		13.0%	-2.7%	-12.3%		13.5%	62.5	29.1	21.6	14.1	10.7	9.1	1.1	1.1	1.1	1.0%	1.6%	1.6%
429A JP	Tekscend	EW	3,740	4,600		23.4%	-20.3%	-8.9%		19.5%	14.2	13.0	11.3	6.3	5.5	4.8	1.9	1.7	1.6	1.9%	1.9%	2.3%
PLAB US Photronics (Not covered)			NA	29	NA	-8.4%	-43.0%	-3.3%		44.6%	15.8	14.5	13.4	NA	NA	NA	NA	NA	NA	NA	NA	NA
Semiconductor Capital Equipment (North America)																						
LRCX US	Lam Research	OW	315	367		83.8%	3.0%	3.1%		125.4%	33.2	26.8	23.2	27.3	22.5	19.8	21.4	15.0	11.9	0.4%	0.4%	0.5%
AMAT US	Applied Materials	EW	480	642		86.8%	11.1%	-10.5%		118.0%	37.5	25.7	19.7	32.3	21.6	17.2	13.9	10.7	8.6	0.4%	0.4%	0.5%
KLAC US	KLA Corp	OW	183	253		50.9%	-2.9%	-12.9%		66.3%	33.4	27.5	24.0	27.6	22.8	20.4	26.1	23.2	15.3	0.5%	0.6%	0.7%
TER US	Teradyne	EW	366	397		89.3%	2.2%	4.7%		135.0%	39.8	31.2	23.7	32.2	24.6	18.4	14.7	11.0	8.4	0.1%	0.1%	0.1%
Mid Small Cap/Manufacturer (Japan):																						
7729 JP	Tokyo Seimitsu	OW	18,010	22,000		62.2%	-1.9%	2.8%		66.2%	23.0	18.7	16.0	13.7	11.6	10.2	3.5	3.1	2.8	1.7%	2.2%	2.5%
6315 JP	Towa	EW	2,337	3,200		8.7%	-19.4%	-18.1%		-7.2%	20.2	15.9	13.6	11.1	9.1	7.8	2.3	2.0	1.8	1.1%	1.2%	1.3%
Semiconductors (Japan)																						
285A JP	Kioxia Holdings	OW	51,260	110,000		391.2%	-21.7%	-8.5%		352.2%	5.0	3.7	3.1	3.4	2.6	2.4	4.5	2.3	1.4	1.2%	2.0%	3.3%

Note: Data as of August 25 market close. Prices and price targets are in Japanese yen except for Teradyne, Applied Materials, Lam Research, KLA and Photronics, which are in US dollars. Tokyo Seimitsu and Towa are covered by Yoshihito Hasegawa. Kioxia Holdings is covered by Kazuo Yoshikawa. Teradyne, Applied Materials, Lam Research, and KLA are covered by Shane Brett; NA = No data found on FactSet/Morgan Stanley Research. Photronics is not covered; Japan Semiconductors: In-Line
Mid Small Cap/Manufacturer (Japan): In-Line, Semiconductor Capital Equipment (North America): In-Line
Source: FactSet, Morgan Stanley Research; Cons = based on FactSet Consensus; MSe = Morgan Stanley Research estimates

Earnings reaction summary for our coverage and overseas peers

	Release date (local time)	result	Actual (LCbn)		vs Cons		Price return (vs Index)	
			Sales	OP	Sales	OP		
Disco	07/23/2026	JunQ	114.31	48.79	+5%	+3%	↓	-11%
Advantest	07/29/2026	JunQ	367.47	189.31	+34%	+67%	↑	+11%
Tokyo Electron	07/30/2026	JunQ	732.39	211.41	+33%	+82%	↑	+5%
Kokusai Electric	08/06/2026	JunQ	75.41	15.89	+22%	+71%	→	-3%
SCREEN Holdings	07/28/2026	JunQ	121.78	14.37	-19%	-54%	↓	-18%
Ebara	08/14/2026	JunQ	244.45	23.84	-17%	-46%	↓	-6%
Lasertec	08/06/2026	JunQ	60.95	27.07	-18%	-25%	↓	-14%
Ulvac	08/14/2026	JunQ	77.50	4.88	+9%	-21%	↓	-16%
Ushio	08/04/2026	JunQ	48.99	4.95	+8%	+33%	↑	+7%
Nikon	08/06/2026	JunQ	164.18	-2.27	-4%	NA	→	-0%
JEOL	08/14/2026	JunQ	24.47	-1.96	-49%	NA	↓	-16%
Tekscend	08/06/2026	JunQ	34.52	7.36	+0%	+5%	↓	-15%
ASML	07/15/2026	JunQ	9.33	3.46	+6%	+9%	→	-1%
Lam Research	07/29/2026	JunQ	6.72	2.51	+26%	+37%	↑	+16%
Applied Materials	08/13/2026	JulQ	9.12	3.08	+30%	+46%	↓	-5%
KLA Tencor	07/28/2026	JunQ	3.66	1.59	+11%	+13%	↓	-9%
Teradyne	07/28/2026	JunQ	1.33	0.44	+23%	+40%	→	-4%

Source: Price return is the index-relative 1-day return after earnings announcements (vs. TOPIX for Japan; vs S&P 500 for US and vs Netherlands AEX for ASML) Source: FactSet, Morgan Stanley Research; Cons = FactSet consensus as of earnings release date

Key SPE makers' WFE market growth outlook for 2026 as of Apr-Jun 2026 results

YoY	Tokyo Electron	Kokusai Electric	SCREEN HD	Lam Research	KLA
WFE outlook	up >= +20%	up >= +25%	up >= +20%	up +36%	up mid-20s%
DRAM	up >= +30%	up +40%+/-	up >= +25%	-	-
NAND	up >= +40%	up +25%+/-	up +20%	-	-
Logic/Foundry	up >= +15%	up +20%+/-	up >= +20%	-	-
China	up +20%	up +5%	> = +10%		

Advanced packaging revenue	up +70%+	2.5x	Flattish	up > +70%	up > +70%
-----------------------------------	----------	------	----------	-----------	-----------

Prev. Q WFE outlook	up >= +20%	up +15%	up +15 ~ 20%	up +27%	up Mid/High teen
DRAM	up >= +30%	up +25 ~ 30%	up +20%- ~ +25%	-	-
NAND	up +40%	up +15%	up close to +15%	-	-
Logic/Foundry	up +10 ~ 15%	up +5 ~ 10%	up > +20 ~ 25%	-	-
China	close to up 20%	down 5%	Flattish	Flat ~ slightly up	flat ~ slightly up

Advanced packaging revenue	up > +60%	flat	up +20% (market growth)	up > +50%	up high-50%
-----------------------------------	-----------	------	----------------------------	-----------	-------------

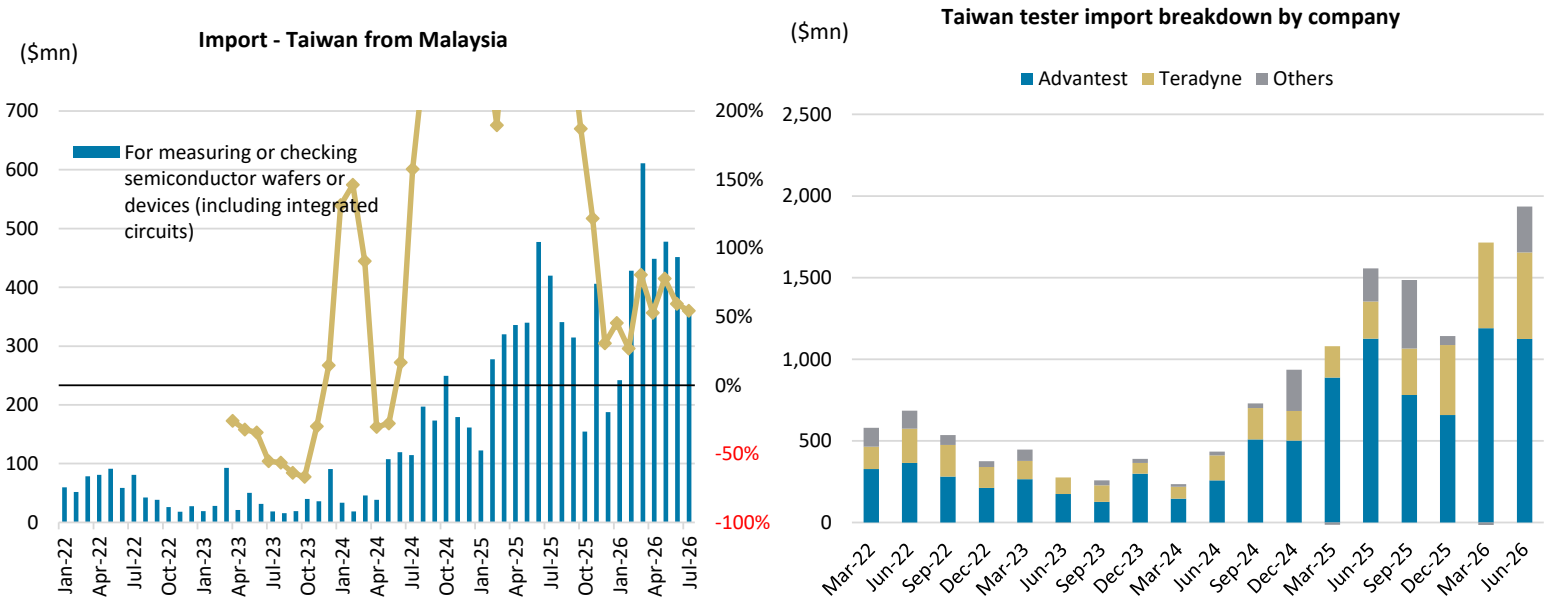
Source: Company comments, Morgan Stanley Research

Capex outlook of major semiconductor manufactures

Capex Forecasts	TSMC	Intel	Samsung Electronics	SK hynix
As of DecQ	\$52B–\$56B (YoY +26%→+36%)	Roughly flat to slightly down (vs. FY25 actual \$18B)	Memory capex to increase (FY25 actual KRW52.2T)	Sharp YoY increase (FY25 actual KRW28.6T)
As of MarQ	\$52B–\$56B, toward the high end of the range (revised up)	Roughly flat YoY (revised up)	Sharp YoY increase (FY25 actual KRW52.2T)	Sharp YoY increase (FY25 actual KRW28.6T)
As of JunQ (new)	\$60B–\$64B (large upward revision, YoY +46%→+55%)	Over \$20B (YoY +13% or more)	NA (policy of continued investment, centered on the DS division)	High KRW40T range (YoY +61%→+71%)
Key drivers of the increase / change	• Rising demand pressure from customers (structural demand including Agentic AI) • Higher equipment purchase costs due to inflation	• Strong customer demand (including long-term agreements) • Equipment investment for Intel 3 / 18A / 18A-P up 40% vs. 2025 • Expanded investment in advanced packaging (EMIB-T)	• Expanded investment in the Pyeongtaek fab and continued strength in AI demand • Ramp-up of the Taylor fab in the U.S.	• Pull-forward of the M15X mass-production schedule • Accelerated investment ahead of the Yongin Fab1 cleanroom opening (early 2027) • Capacity expansion to meet AI demand
Medium-term capex policy from 2027 onward	Stated that "capex over the next three years will significantly exceed the past three years (= \$101B in total)" Announced an additional \$100B investment in Arizona (cumulative \$265B), with four or more additional fabs planned	2027 capex expected to be significantly above the 2026 level	Construction of Taylor Fab2 to start at end-2026, with mass production targeted for 2030	Domestically, strengthening Icheon and Yongin as core DRAM / AI memory sites and Cheongju as the hub for NAND and advanced packaging
Main investment targets / regions	• Taiwan: leading-edge fabs and advanced packaging fabs (13 buildings under construction) • Arizona: 2nm-and-below logic plus advanced packaging (cumulative \$265B) • Japan: new fab to be ramped up early	• U.S. accounts for the majority of investment • Equipment investment for Intel 3, 18A and 18A-P • Advanced packaging (EMIB-T)	• Pyeongtaek: expanded fab investment • Taylor: foundry Fab1 ramp-up (operation planned within 2026), Fab2 groundbreaking (end-2026, mass production targeted 2030)	• Centered on Yongin cluster infrastructure development and the M15X ramp-up • Announced investment plans for advanced packaging (P&T7) and a new NAND production site (M17)

Source: Company comments, Morgan Stanley Research

Taiwan test-related equipment imports from Malaysia



Source: Ministry of Economic Affairs, R.O.C., Morgan Stanley Research

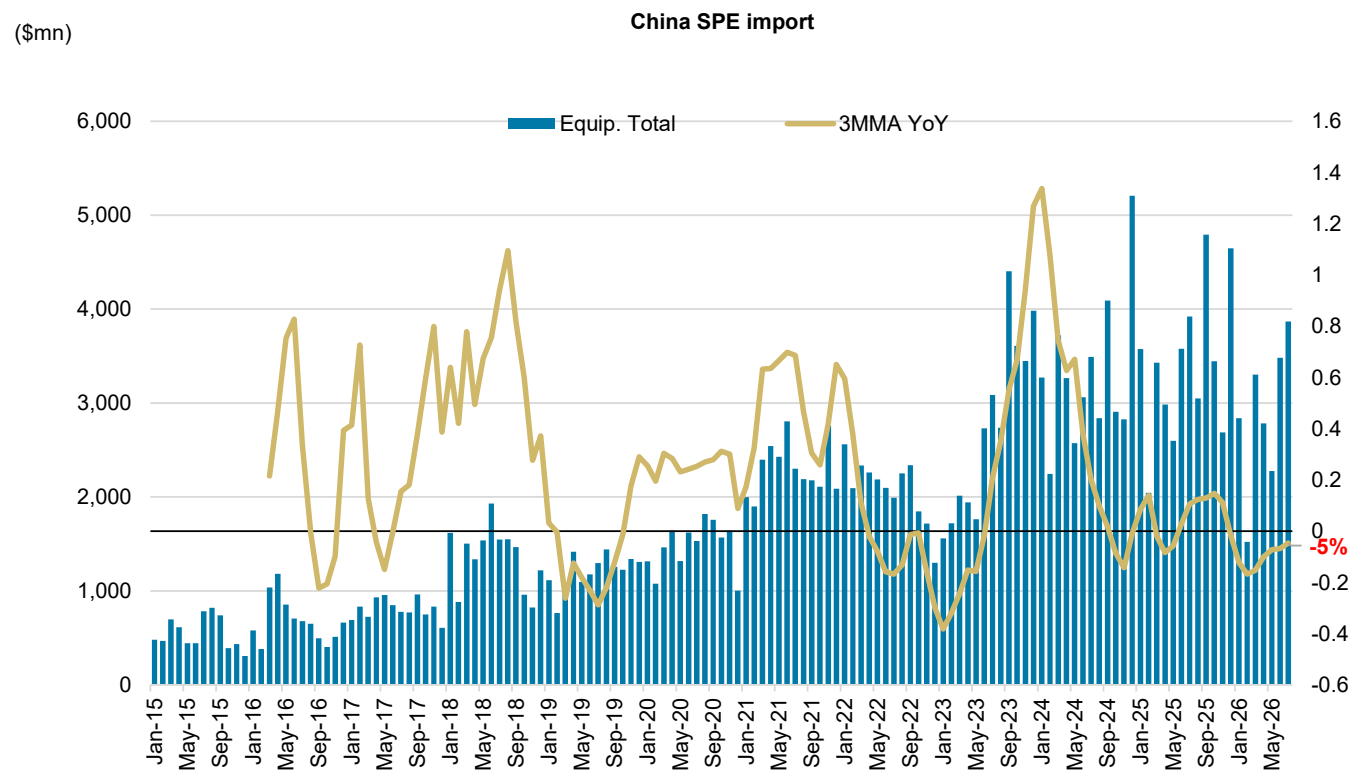
Quarterly system sales of major front-end equipment manufacturers

(\$mn) Equipment sales	CY2025				CY2026		CY2026	CY2025
	Q1	Q2	Q3	Q4	Q1	Q2	YTD prog.	
Total	21,464	21,636	21,356	23,183	24,055	25,645	49,701	87,639
y-y	26%	22%	6%	4%	12%	19%	57%	13%
q-q	-4%	1%	-1%	9%	4%	7%	(prog. vs 2025)	
ASML	6,211	6,591	6,520	8,910	7,258	7,497	14,754	28,232
Applied Materials	5,401	5,564	4,879	5,141	5,965	7,040	13,005	20,985
Lam Research	3,035	3,438	3,548	3,357	3,731	4,250	7,980	13,378
Tokyo Electron	3,360	2,742	3,077	2,459	3,403	3,268	6,671	11,639
KLA Tenchor	2,394	2,472	2,465	2,511	2,640	2,837	5,477	9,842
SCREEN Holdings	750	609	586	605	740	395	1,135	2,549
KOKUSAI Electric	313	220	281	200	318	359	677	1,015
Logic/Foundry	10,825	12,024	11,858	13,108	11,664	12,585	24,249	47,814
y-y	27%	29%	4%	14%	8%	5%	51%	17%
q-q	-6%	11%	-1%	11%	-11%	8%	(prog. vs 2025)	
DRAM	6,675	5,131	5,709	6,920	8,576	8,224	16,800	24,434
y-y	27%	-1%	5%	1%	28%	60%	69%	8%
q-q	-3%	-23%	11%	21%	24%	-4%	(prog. vs 2025)	
NAND	1,612	2,005	1,625	1,047	1,171	2,003	3,174	6,288
y-y	64%	167%	147%	-20%	-27%	-0%	50%	70%
q-q	24%	24%	-19%	-36%	12%	71%	(prog. vs 2025)	
Sales to China	7,737	8,983	10,464	9,711	8,026	7,887	15,913	36,894
y-y	-21%	-7%	4%	7%	4%	-12%	43%	-4%
q-q	-15%	16%	16%	-7%	-17%	-2%	(prog. vs 2025)	

Note: Total represents aggregate new system sales of ASML, AMAT, LRCX, TEL, KLAC, SCREEN, and KOKUSAI listed in the table; application-level figures represent the sum of sales by application across the respective companies. Application classifications are based on company disclosures and therefore the sum may not reconcile with Total. ASML Memory sales are classified as DRAM for simplicity. YTD represents CY26 YTD sales; prog. vs. 2025 represents CY26 YTD sales as a percentage of CY25 full-year sales. FX conversion is based on Visible Alpha methodology.

Source: Visible Alpha, Morgan Stanley Research

China monthly SPE import data: Jul -5% YoY, 3MMA



Note: Data excludes handling equipment and components

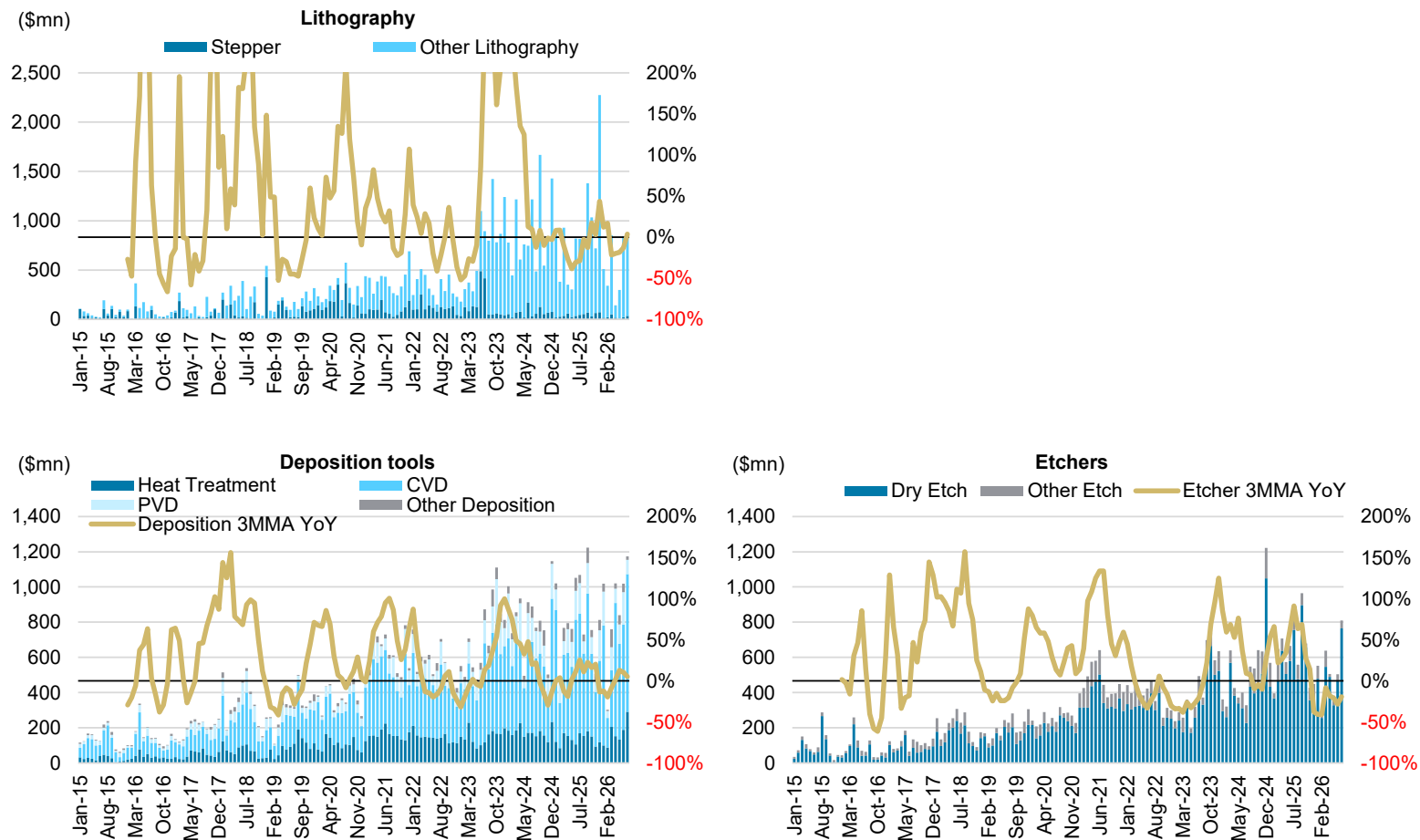
Source: General Administration of Customs of the People's Republic of China, Morgan Stanley Research

WFE forecast table

(\$mn)	2022	2023	2024	2025	2026e	2027e	2028e
WFE Revenue	96,222	96,008	105,057	117,530	162,782	223,360	254,124
y-y	6%	0%	9%	12%	39%	37%	14%
WFE by segment							
Foundry/Logic	59,180	70,023	69,192	75,307	94,818	127,995	143,672
Leading Logic		27,500	24,500	31,000	51,189	76,572	89,589
Mature Logic		42,523	44,692	44,307	43,629	51,422	54,083
Memory	35,854	24,794	35,228	41,600	66,515	94,048	108,722
DRAM	18,256	19,579	29,244	31,083	52,827	70,086	77,456
NAND	17,598	5,215	5,985	10,517	13,689	23,963	31,266
Other	1,188	1,192	637	622	1,449	1,316	1,730
Y/Y Change %							
Foundry/Logic	18%	18%	-1%	9%	26%	35%	12%
Leading Logic			-11%	27%	65%	50%	17%
Mature Logic			5%	-1%	-2%	18%	5%
Memory	-10%	-31%	42%	18%	60%	41%	16%
DRAM	-12%	7%	49%	6%	70%	33%	11%
NAND	-8%	-70%	15%	76%	30%	75%	30%
Other	19%	0%	-47%	-2%	133%	-9%	31%
WFE by Region							
North America	10,239	11,347	13,657	10,578	15,464	22,336	25,412
Europe	5,982	5,742	4,202	2,351	2,767	3,350	3,812
Japan	7,244	6,439	7,354	8,815	11,395	16,082	18,551
Korea	19,656	18,966	18,910	23,506	39,068	53,606	59,211
Taiwan	26,566	19,501	15,233	27,620	43,951	62,541	71,155
China	22,104	31,681	42,548	41,723	45,579	55,840	68,613
Other	4,431	2,334	3,152	2,938	4,558	9,604	7,370
WFE by Region (Y/Y)							
North America	32%	11%	20%	-23%	46%	44%	14%
Europe	108%	-4%	-27%	-44%	18%	21%	14%
Japan	5%	-11%	14%	20%	29%	41%	15%
Korea	-14%	-4%	0%	24%	66%	37%	10%
Taiwan	13%	-27%	-22%	81%	59%	42%	14%
China	-10%	43%	34%	-2%	9%	23%	23%
Other	73%	-47%	35%	-7%	55%	111%	-23%

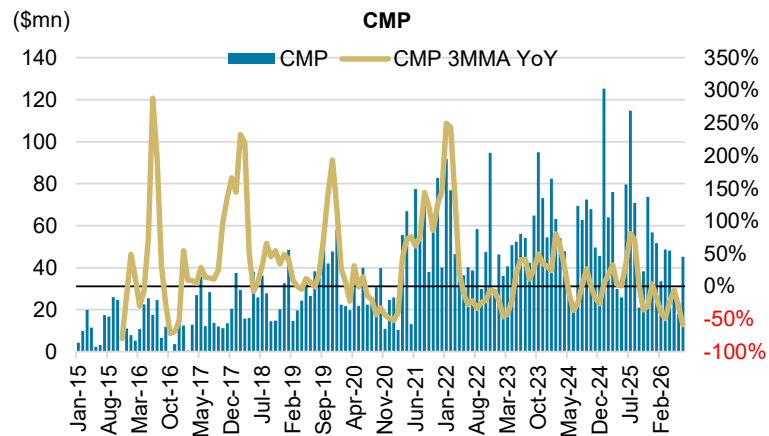
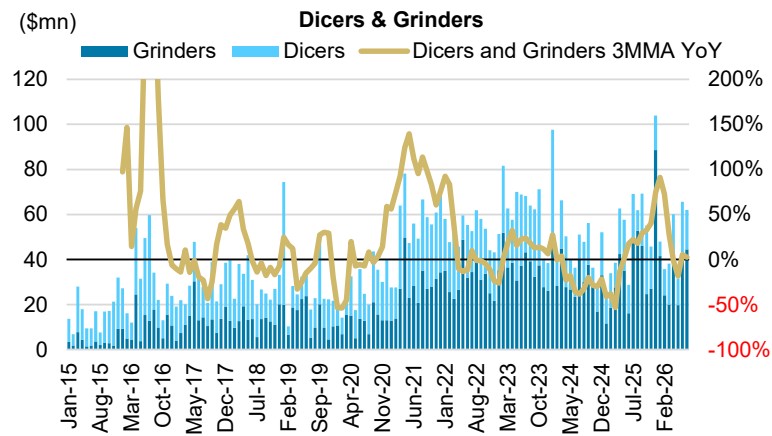
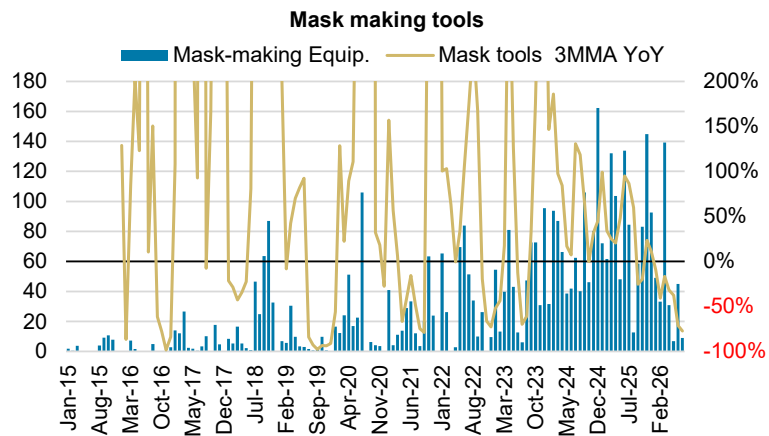
Source: Gartner, SEMI, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

China monthly SPE import data



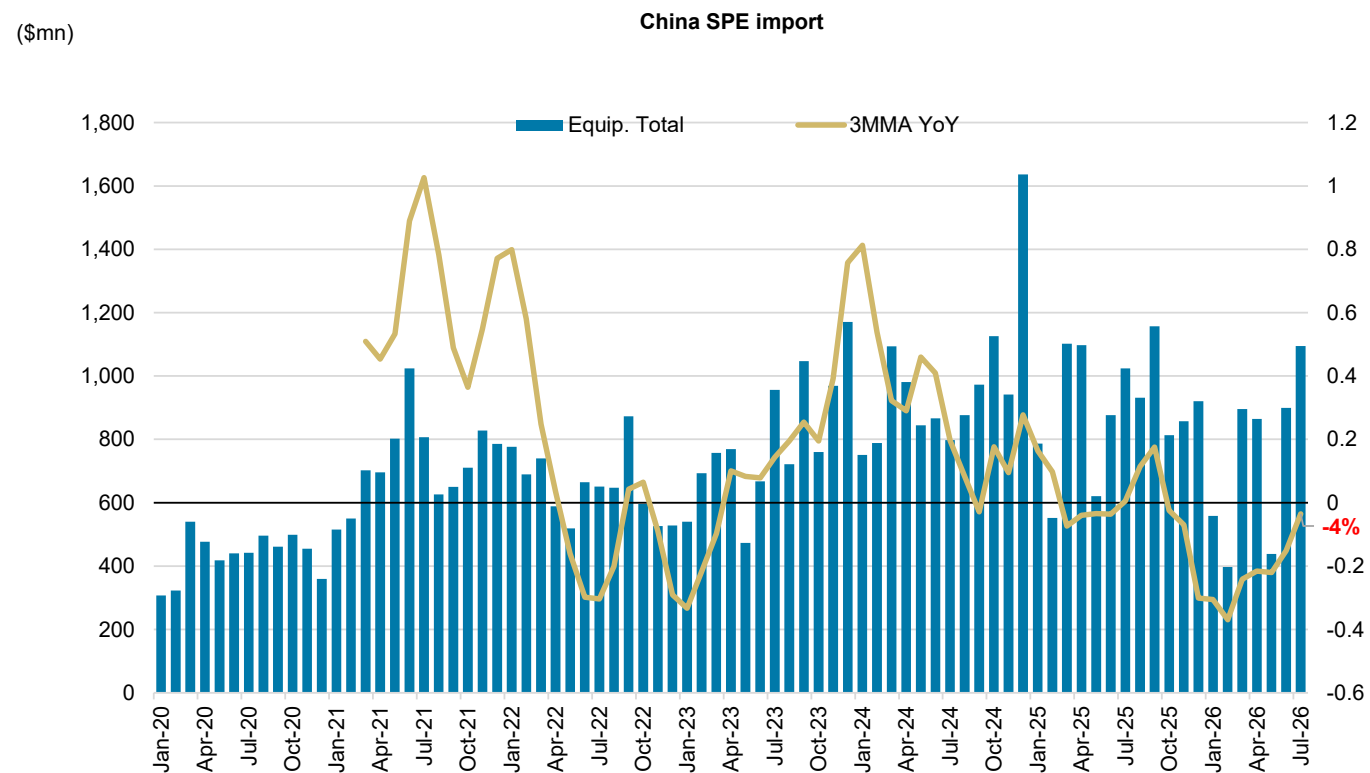
Source: General Administration of Customs of the People's Republic of China, Morgan Stanley Research

China monthly SPE import data



Source: General Administration of Customs of the People's Republic of China, Morgan Stanley Research

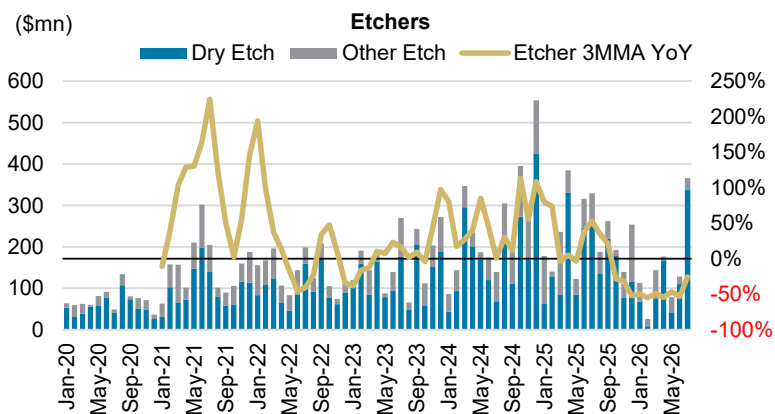
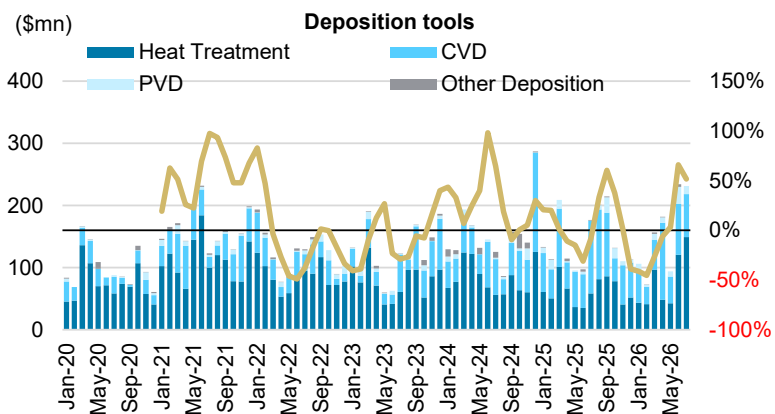
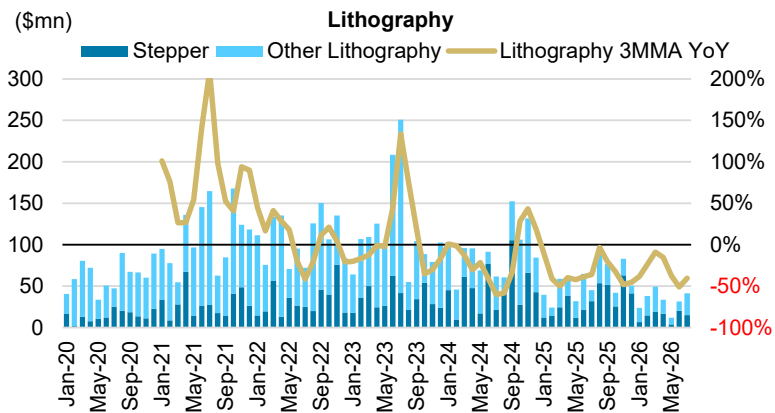
China monthly SPE imports from Japan: Jun -4% YoY, 3MMA



Note: Data excludes handling equipment and components

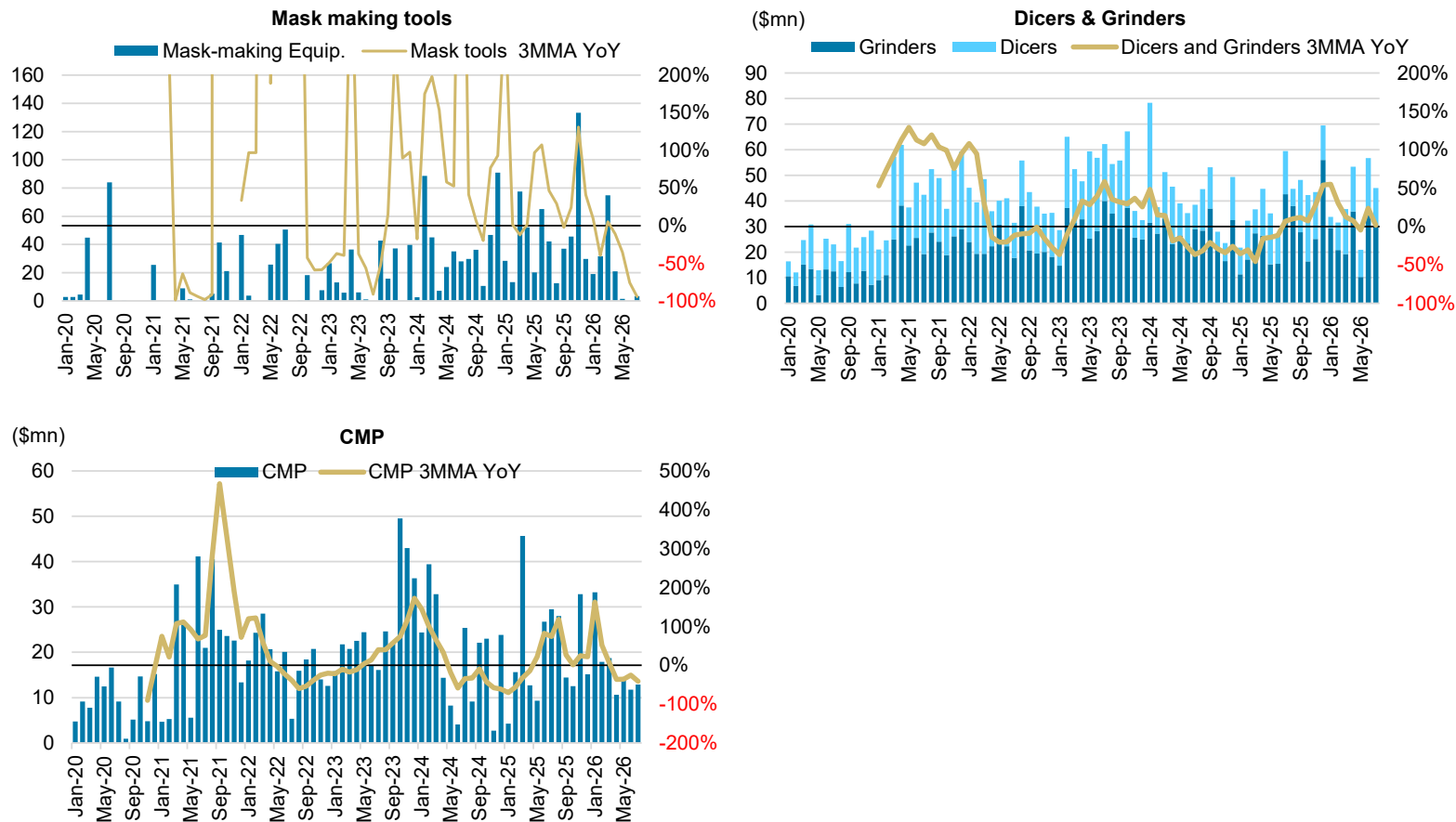
Source: General Administration of Customs of the People's Republic of China, Morgan Stanley Research

China monthly SPE imports from Japan



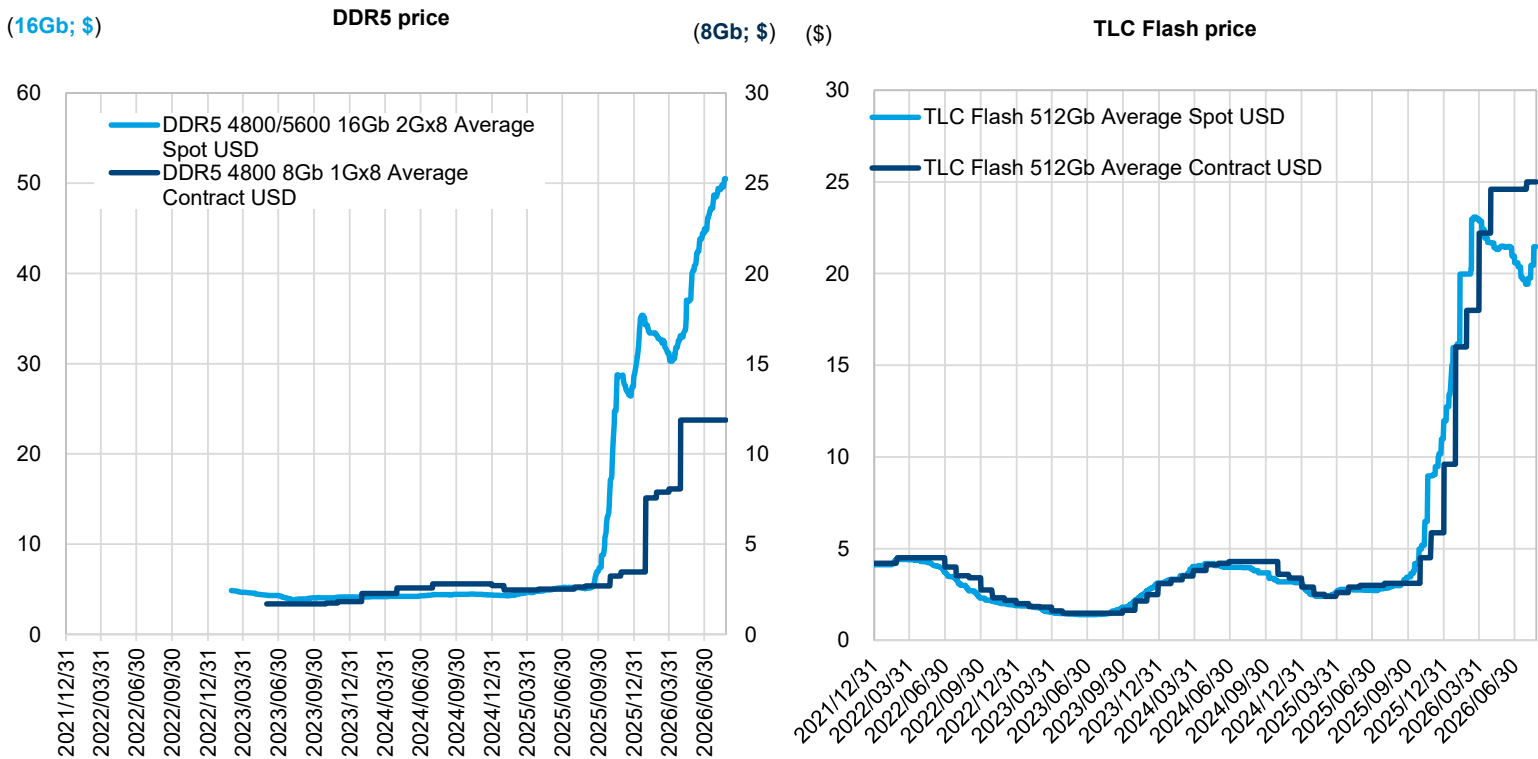
Source: General Administration of Customs of the People's Republic of China, Morgan Stanley Research

China monthly SPE imports from Japan



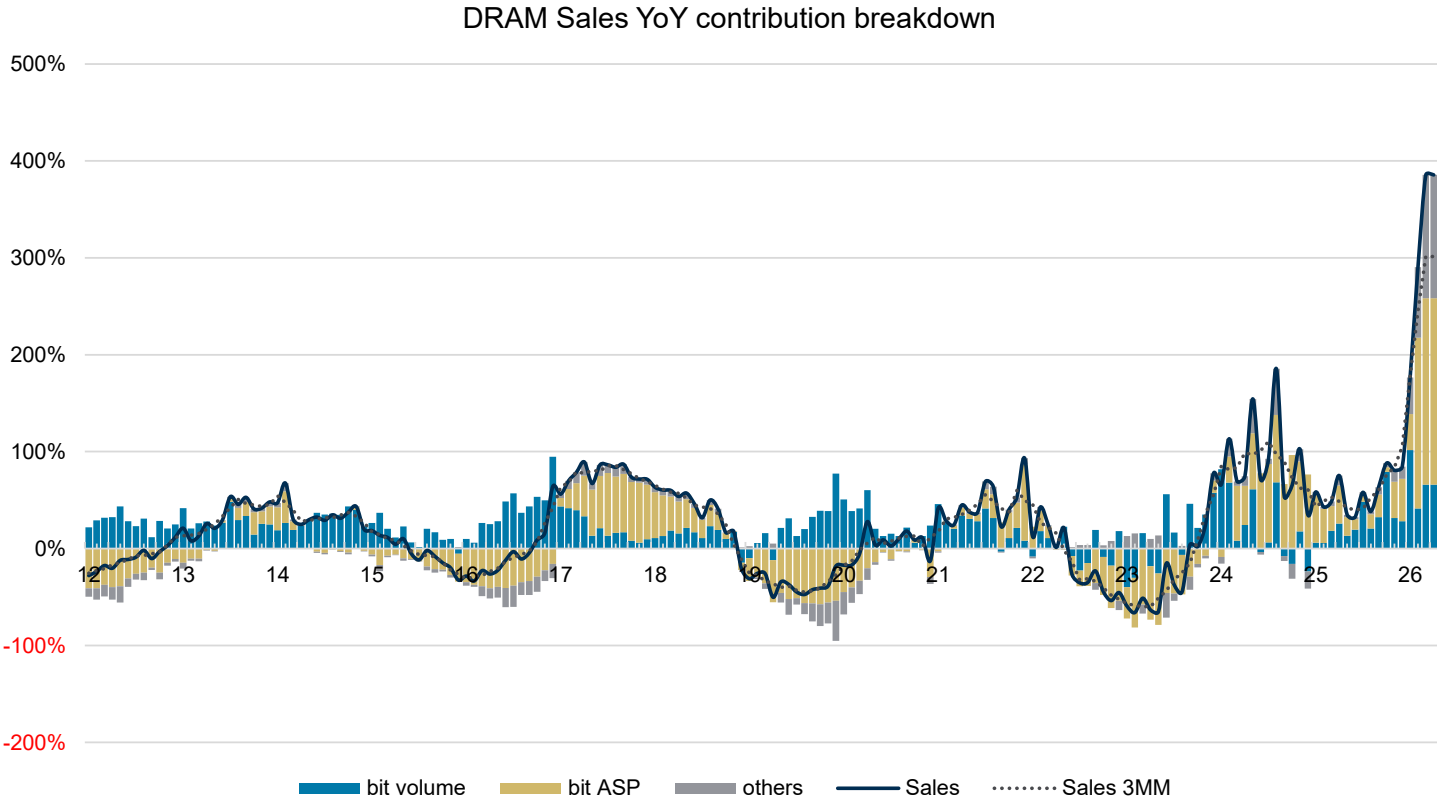
Source: General Administration of Customs of the People's Republic of China, Morgan Stanley Research

Memory price trends



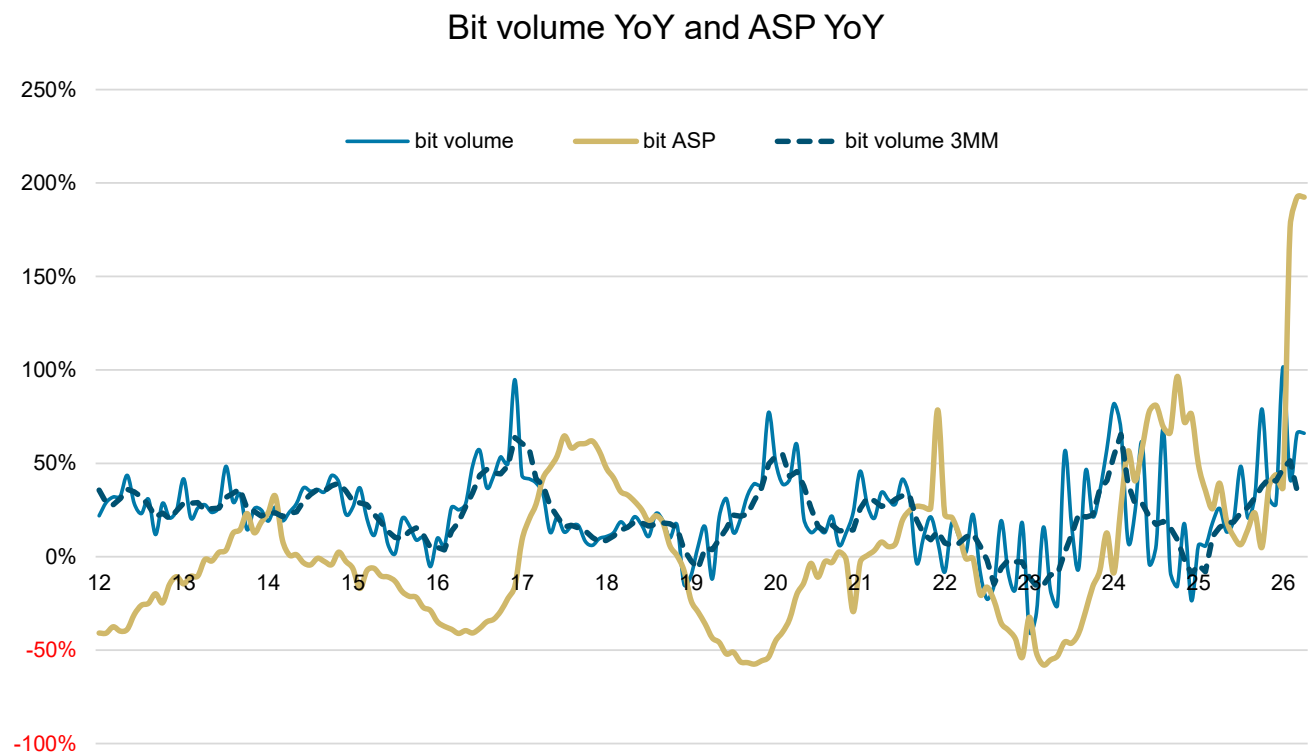
Source: FactSet, Morgan Stanley Research

DRAM YoY growth



Source: WSTS, Morgan Stanley Research

DRAM bit shipment and ASP YoY growth



Source: WSTS, Morgan Stanley Research

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared by Morgan Stanley MUFG Securities Co., Ltd. and its affiliates (collectively, "Morgan Stanley").

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1800 303-2495; Hong Kong +852 2848-5999; Latin America +1718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Suzune Tamura, CFA; Kazuo Yoshikawa, CFA.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

As of July 31, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: Advantest, Applied Materials Inc., DISCO, Ebara, KIOXIA Holdings, KLA Corp, KOKUSAI ELECTRIC, Lam Research Corp, Lasertec, Nikon, Teradyne Inc, Tokyo Electron, Tokyo Seimitsu, Ulvac, Ushio.

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of KOKUSAI ELECTRIC, Tekscend Photomask.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Applied Materials Inc., JEOL, Tekscend Photomask, Tokyo Electron.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from Advantest, Applied Materials Inc., DISCO, Ebara, JEOL, KIOXIA Holdings, KLA Corp, KOKUSAI ELECTRIC, Lam Research Corp, Lasertec, Nikon, SCREEN Holdings, Tekscend Photomask, Teradyne Inc, Tokyo Electron, Tokyo Seimitsu, Towa, Ulvac, Ushio.

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from Applied Materials Inc., Tokyo Electron.

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Advantest, Applied Materials Inc., Ebara, JEOL, KIOXIA Holdings, KLA Corp, KOKUSAI ELECTRIC, Lam Research Corp, Lasertec, Nikon, SCREEN Holdings, Tekscend Photomask, Teradyne Inc, Tokyo Electron, Ulvac, Ushio.

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: Applied Materials Inc., Tokyo Electron.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of July 31, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1529	42%	442	47%	29%	750	43%
Equal-weight/Hold	1582	43%	396	42%	25%	773	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	560	15%	97	10%	17%	217	12%
Total	3,674		936			1741	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to <https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch>.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act. Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients

through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>. Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research.

Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management. Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V. which is regulated by Comisión Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM00001203), and depository participant with National Securities Depository Limited

(SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

INDUSTRY COVERAGE: Semiconductor Production Equipment

COMPANY (TICKER)	RATING (AS OF)	PRICE* (08/25/2026)
Suzune Tamura, CFA		
Advantest (6857.T)	O (01/17/2024)	¥34,450
DISCO (6146.T)	O (07/07/2023)	¥60,700
Ebara (6361.T)	O (12/04/2024)	¥4,931
JEOL (6951.T)	E (07/08/2025)	¥7,140
KOKUSAI ELECTRIC (6525.T)	O (12/19/2025)	¥8,489
Lasertec (6920.T)	U (07/28/2025)	¥34,950
Nikon (7731.T)	U (08/08/2023)	¥1,971
SCREEN Holdings (7735.T)	O (11/13/2025)	¥13,305
Tekscend Photomask (429A.T)	E (11/18/2025)	¥3,740

Tokyo Electron (8035.T)	O (12/19/2025)	¥55,410
Ulvac (6728.T)	E (01/17/2024)	¥7,597
Ushio (6925.T)	O (01/05/2026)	¥3,900

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

© 2026 Morgan Stanley MUFG